

Antibiotics

Cell Wall Inhibitors

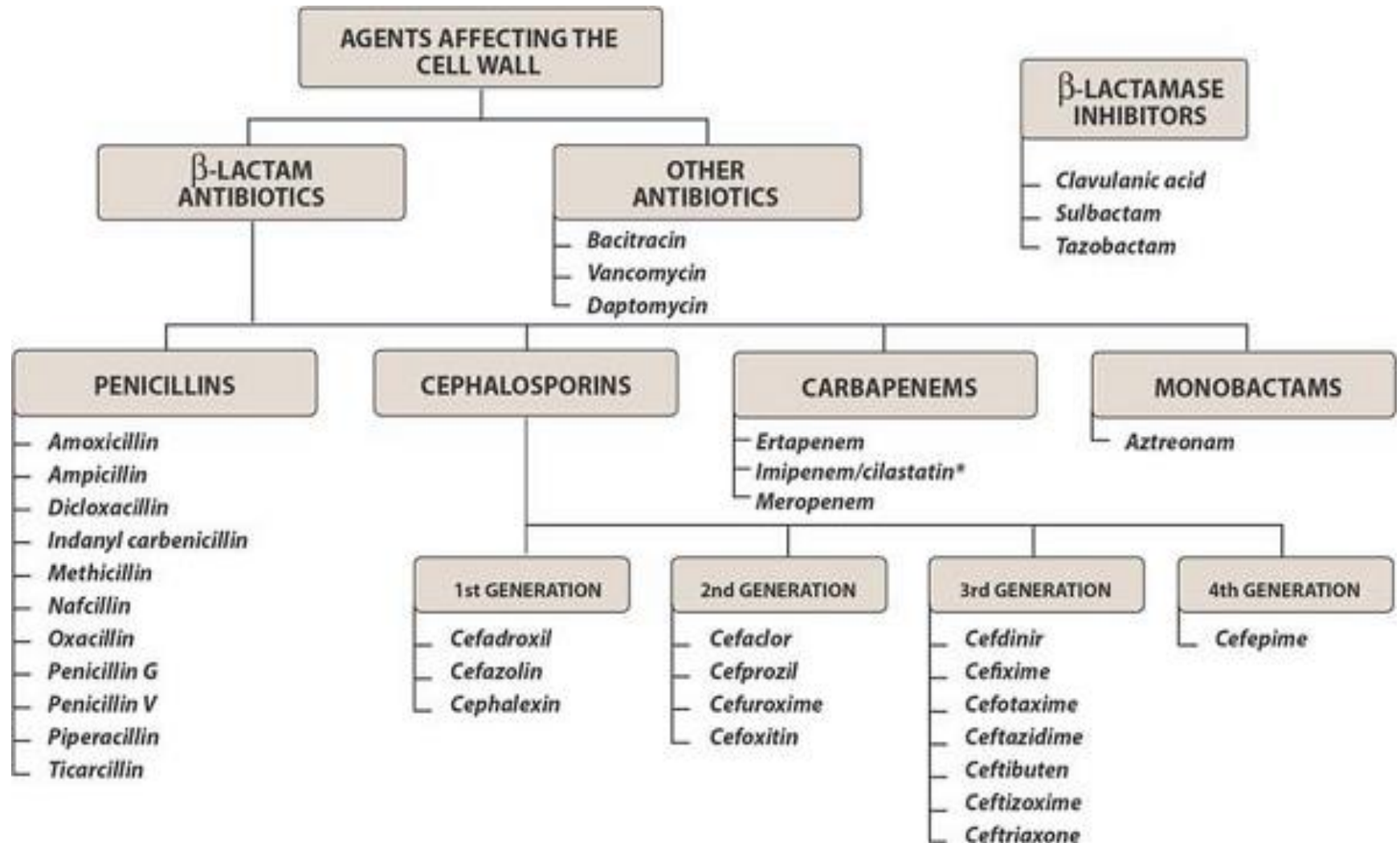


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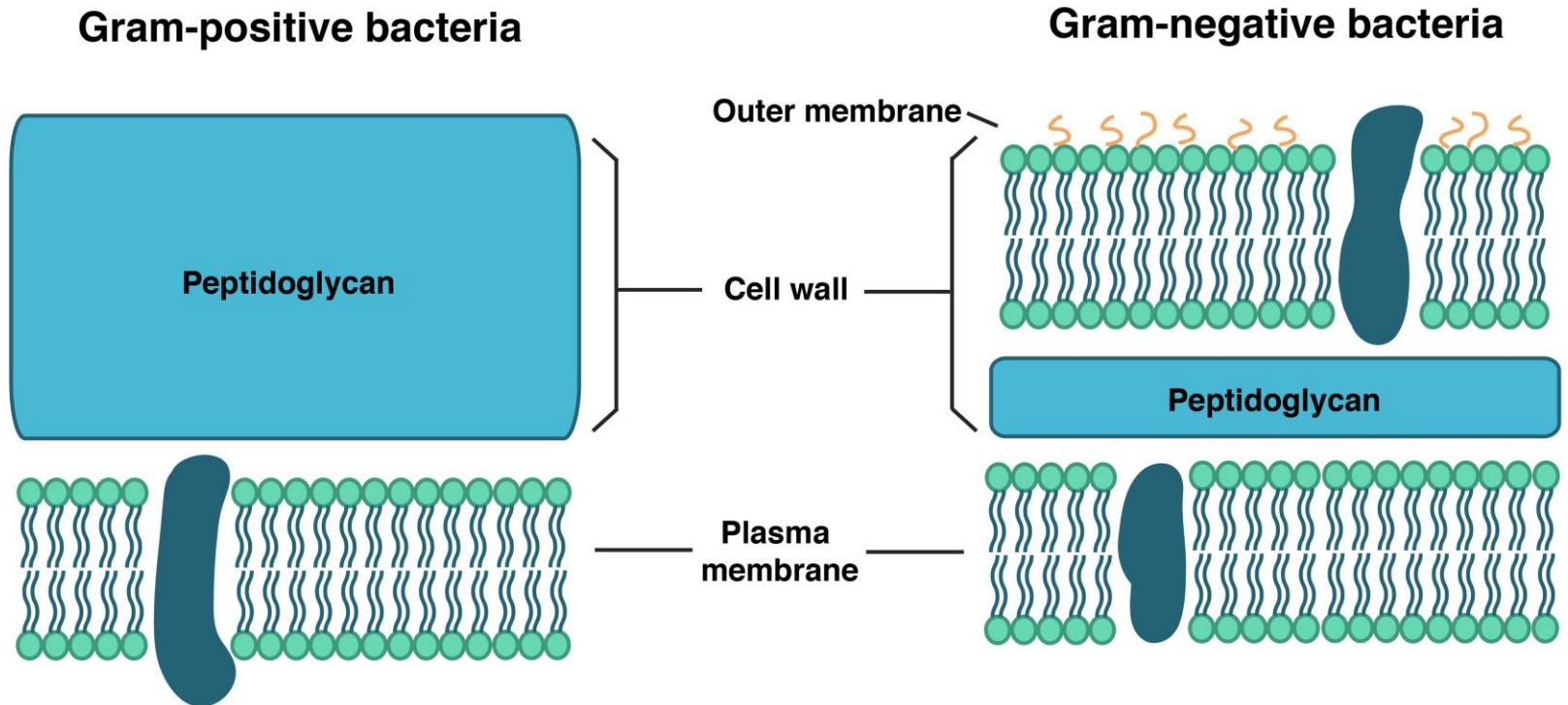
Intended learning outcomes (ILOs)



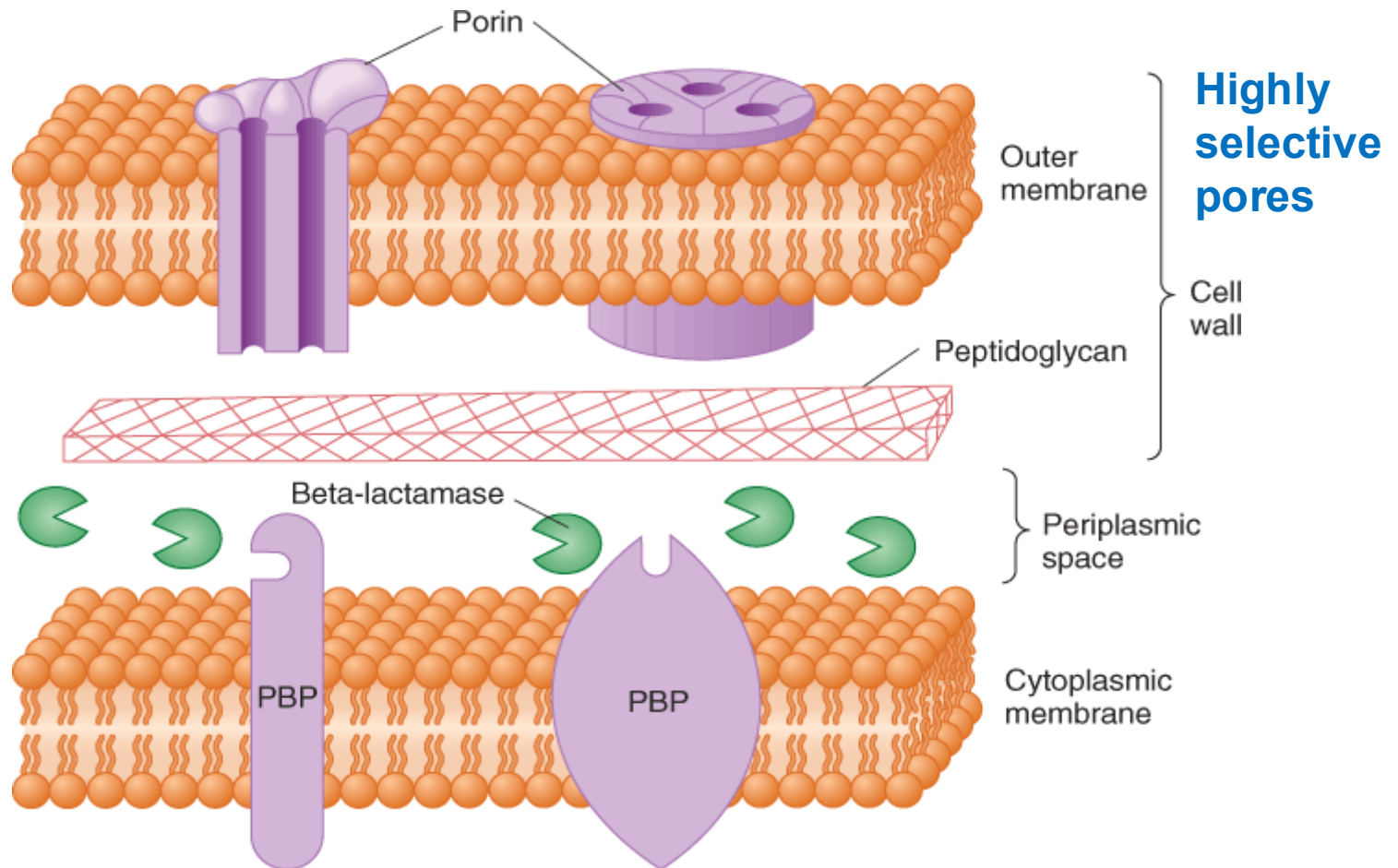
- How do cell wall inhibitors work?
- Discuss their pharmacokinetics
- Apply the pharmacological effects of antimicrobial drugs for its major clinical uses and discuss their adverse effects
- Define the actions and uses of **Vancomycin**
- Why do you **always** have to **finish** your prescription?



Bacterial cell wall structure



Bacterial cell wall structure



Source: Trevor AJ, Katzung BG, Masters SB: *Pharmacology Examination & Board Review, 9th Edition*: www.accesspharmacy.com
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PBP: penicillin binding peptide

Resistance against Penicillins

↓
Natural (Innate Resistance)

Lack a peptidoglycan cell wall

- Fungi
- Virus
- Human cells
- Mycoplasma
- Mycobacteria
- Chlamydia

Has peptidoglycan cell wall but still resistant

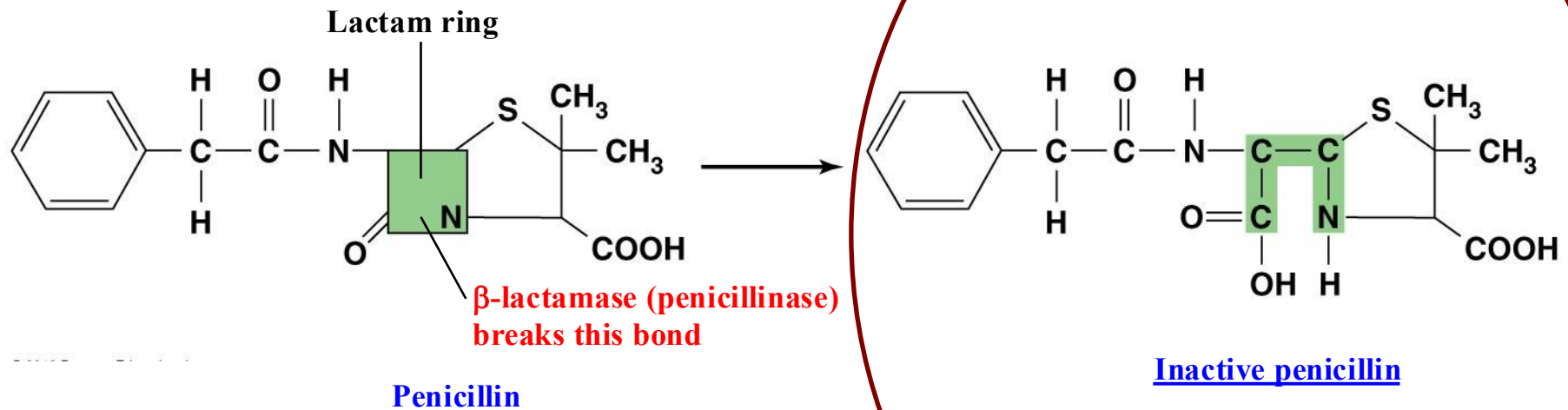
- *P. aeruginosa* ≠ PenG



↓
Acquired

(Initially susceptible then later got resistant)

- 1- β -lactamase production
(plasmid-mediated β -lactamase)
- 2- Altering the porin (↓ permeability to drug)
- 3- Increase efflux pump
- 4- Alteration of PBP (MRSA, VRSA)
(*S. pneumoniae* ≠ PenG)



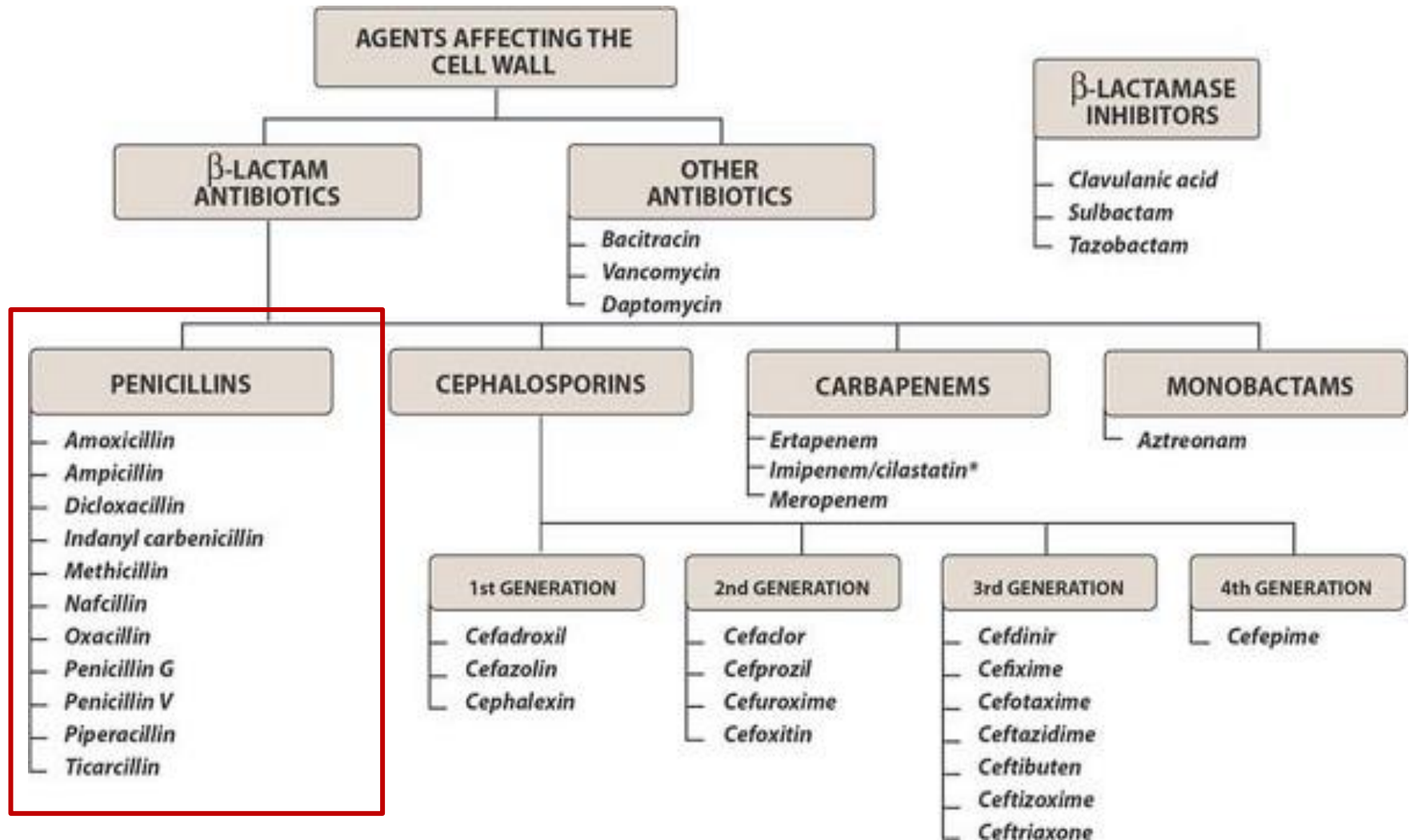
How beta-lactamase (penicillinase) renders penicillin inactive

Antibiotics acting on cell wall of bacteria



- Beta lactams:
(Penicillins, cephalosporins, carbapenems, monobactam)
- Glycopeptides:
(Vancomycin, teicoplanin)

Bacterial cell wall synthesis inhibitors



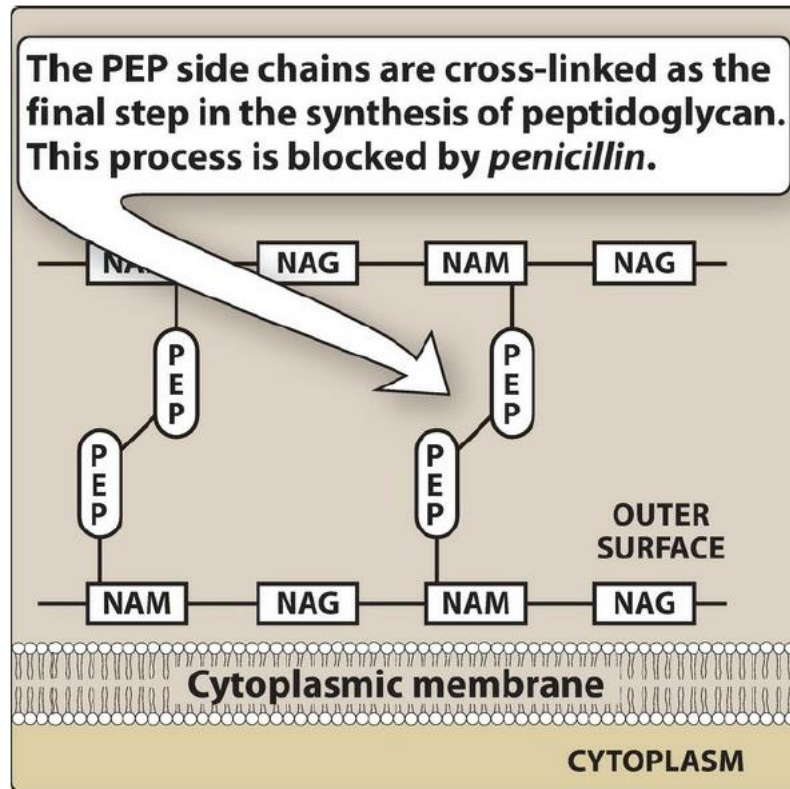
Penicillins: Mechanism of Action

Penicillins enter the bacteria via the cell wall

- Inside the cell, they bind to penicillin-binding-protein (PBP)
(transpeptidase enzyme responsible for peptidoglycan cross-linkings)
- Once bound, normal cell wall synthesis is disrupted.
- Bacteria become hyperosmotic/ love to absorb water
- Bacteria swelling then burst.
- These drugs are bactericidal.
- Active against actively proliferating and not resting bacteria.
- Inactive against protozoa, mycobacteria, fungi and viruses.



Penicillins: Mechanism of Action



PEP: cross-linking peptides

PENICILLINS

Natural Pens.

v.v. Na~~RR~~ow spectrum

Extended-spectrum Pens.

Antipseudomonal Pens.

(Benzyle Penicillins)

(Antistaphylococcal Pens.)

(Aminopenicillins)

Pen G

Pen V

I.V

P.O.

• G+ cocci

• G+ bacilli

• G- cocci

(Gonococcus)

• Spirochetes

(T. pallidum)

(Syphilis)

- Methicillin
- Cloxacillin
- Floxacillin
- Oxacillin
- Dicloxacillin
- Nafcillin

- Ampicillin (I.V)
- Amoxicillin (P.O.)
- Ampicillin +
Sulbactam = Unasyn
- Amoxicillin +
clavulanic acid
= Augmentin

- Active against G+ and G-
- More active against G- organisms.

- Carbenicillin
- Ticarcillin +
clavulanic acid =
Timentin
- Piperacillin +
tazobactam = Zosyn

- A active against P. aeruginosa
- For Pseud. septicemia, they should be taken together with an aminoglycoside.



	PENICILLIN G	ANTI-STAPH PENICILLINS	EXTENDED SPECTRUM PENICILLINS	ANTI- PSEUDOMONAL PENICILLINS
		Cloxacillin Flucloxacillin Nafcillin	Ampicillin Amoxicillin	Ticarcillin Azlocillin Pipracillin
Spectrum	Gram +ve spp. and few Gram -ve cocci <i>T. pallidum</i>	<i>Staphylococci</i>	Gram +ve and Gram -ve spp. (except <i>Pseudomonas</i>)	<i>Pseudomonas</i> and Gram -ve spp.
β-lactamase susceptibility	Yes	No	Yes	Yes
			They are combined with clavulanic acid or sulbactam to cover β-lactamase producing bacteria.	

- **Long-acting penicillins:**
- Some Penicillins are used as **Depot form (I.M)**; active penicillins release slowly (Given in upper outer part of Buttocks, children; thigh)

Procaine Penicillin G (12-24 h)

Benzathine Penicillin G (Rheumatic fever) (3-4 Weeks)

Used as Prophylaxis:

- 1- **Against streptococcus pyogens in patients with Rheumatic fever**
- 2- **Patients with artificial valves (endocarditis)**
- 3- **After delivery (newborn with gonorrheal conjunctivitis)**

β- Lactamase Inhibitors

Formulations with B-lactamase inhibitors protect amoxicillin or ampicillin from enzymatic hydrolysis and extends their antimicrobial spectrum.:

Clavulanic acid + amoxicillin (**Augmentin**) – Oral.

Sulbactam + ampicillin (**Unasyn**) (IV-IM).





Pseudomonas Euroginosa

Penicillins: Pharmacokinetics

➤ Absorption:

- Most of the penicillins are poorly absorbed orally Except:

Penicillin V

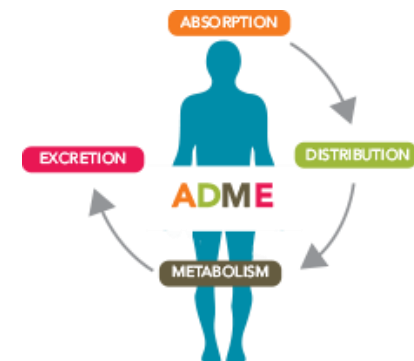
Amoxicillin

Amoxicillin + clavulanic acid = Augmentin®

Indanyl Carbenicillin

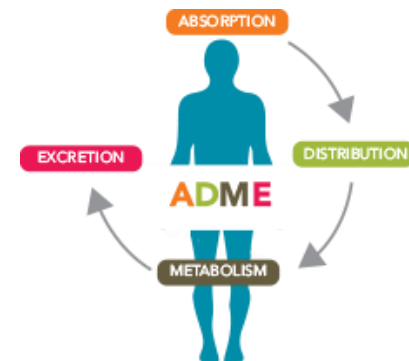
- Given in empty stomach (1 h before or 2 h after food)

- Amoxicillin is absorbed mostly from GIT
- Ampicillin is very poor oral absorption



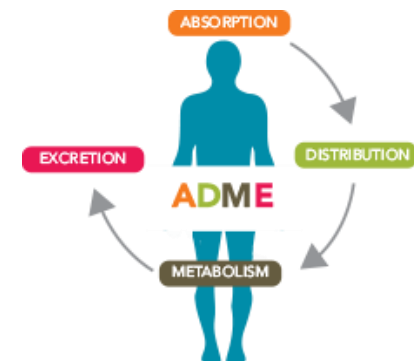
➤ Distribution:

- **Not** lipid soluble to pass through cell membrane; not good for intracellular bacteria.
- **Penicillins normally can not cross BBB (Ionized molecules); only in inflammation (e.g. Meningitis) Penicillins can cross into CSF and cross bone (osteomyelitis)**
- If cross Placenta- **not teratogenic**
- **The safest Antibiotics to be given to pregnant women**



➤ Metabolism:

- They are very safe drugs, no need for monitoring.
- Excreted mainly unchanged in urine by active tubular secretion (90%) (except. Nafcillin, cloxacillin, dicloxacillin and oxacillin; through liver)
- $T_{1/2}$ =30-60 min (10 h in renal failure)
- **Probenecid** bind with organic acid transport mechanism in the kidney and inhibits Penicillins tubular secretion, can increase their blood levels
- Penicillins appear in milk
- Excreted in saliva



➤ Penicillins : clinical uses

- Streptococcal infections (throat infections, bacterial endocarditis *in patients having dental surgery*)
- Staphylococcal Infections
- Spirochetes (Syphilis) and Gonorrhoea (Gram G- cocci)
- Rheumatic Fever (**Benzathin Pen G**)
- Diphtheria
- Salmonella infections

- *Penicillins + Aminoglycosides = Synergistic effects*
- **Prophylactic** - in patients with congenital heart disease
 - Rheumatic fever



➤ Adverse effects

1. Type-1 hypersensitivity reaction (IgE-immediate reactions) (any immune reaction which is damaging the tissues (occur in 5-10% of patients))
 - Most dangerous is **anaphylactic shock** (1-3:10.000 person)
2. Diarrhoea (**Pseudomembranous colitis** by *c.difficile* especially with **Ampicillin**) :Treated with
 - **Metronidazole orally or Vancomycin orally too.**
3. Hypovitaminosis (Vit K)
4. Platelete dysfunction (especially with Piperacillin, Nafcillin)
5. May cause **seizures** (**Not used intrathecally**); also inhibit GABA especially procaine penicillin

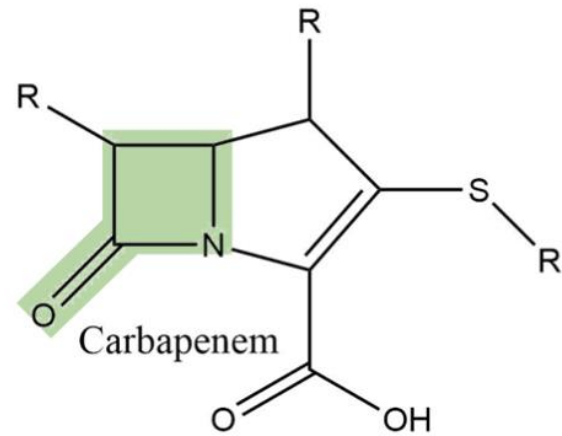
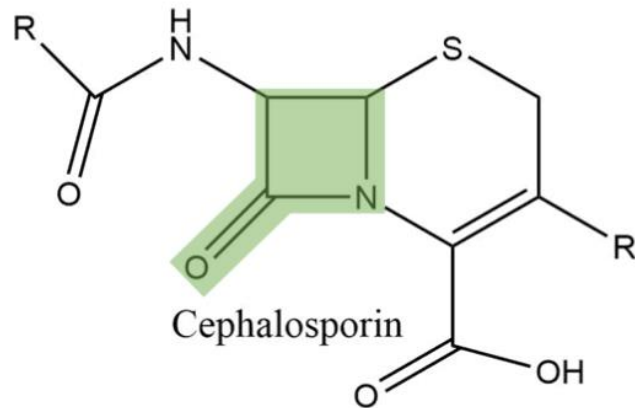
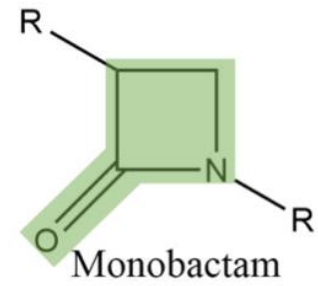
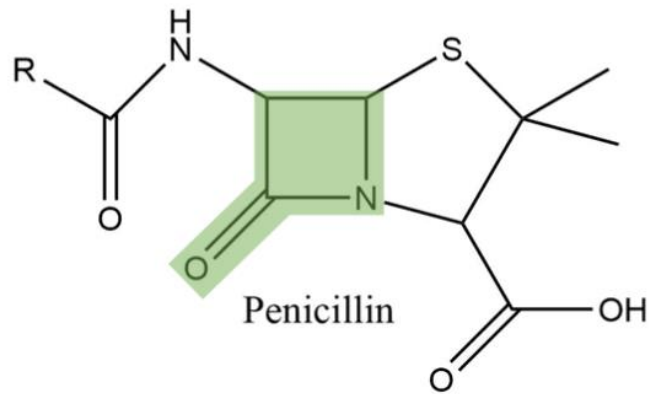


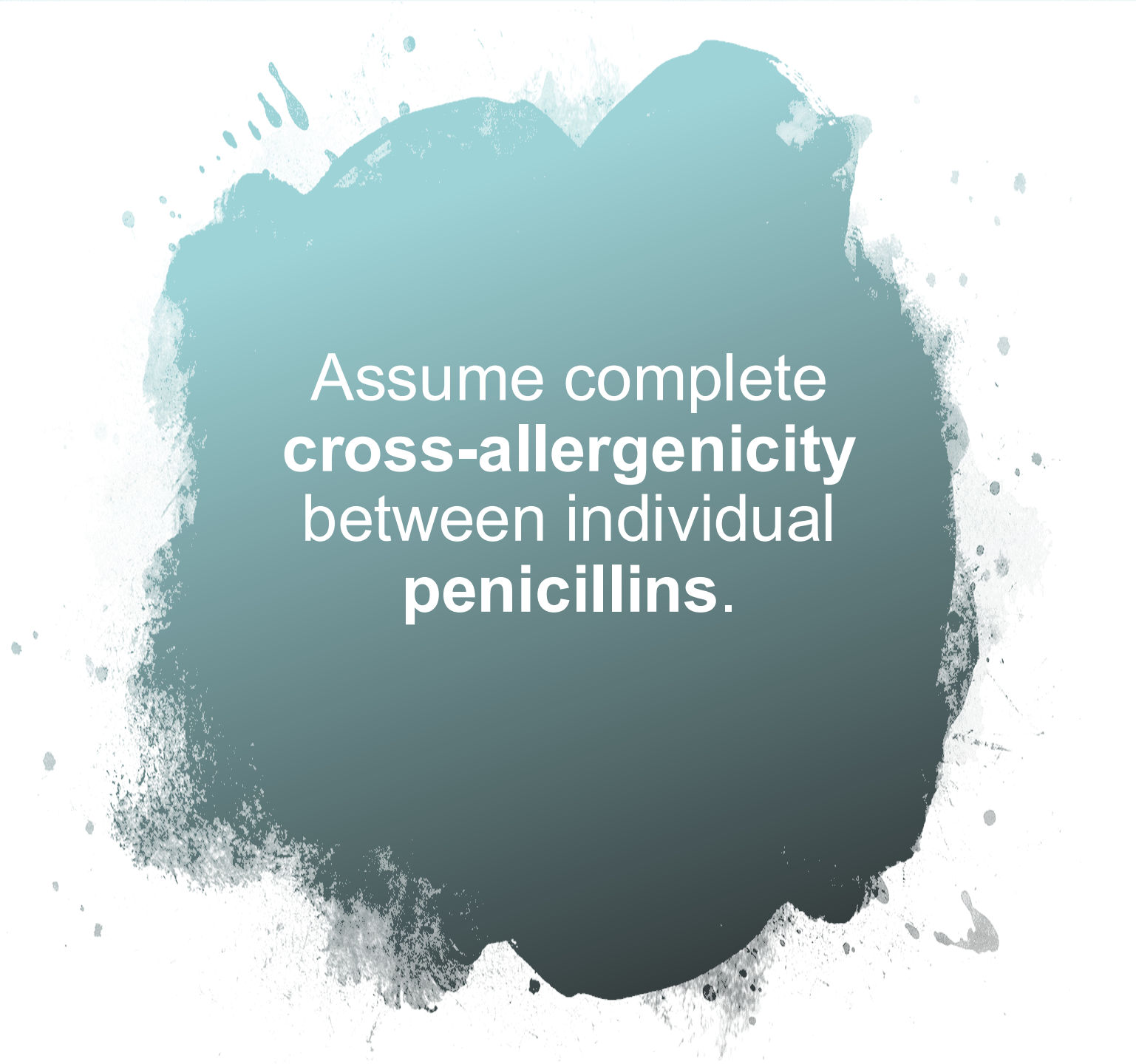
Antibiotic skin testing (AST)

- For identifying patients with IgE-mediated hypersensitivity reactions to antibiotics.
- **About 5-10% hypersensitivity**
- **1:10,000 patients (anaphylactic shock)**
- Performed on the **volar forearm** using the **intradermal method**.
- The injection amounts varied among the hospitals, **ranging from 10 µl**
- The time lag (**15 to 20 minutes**).
- **Wheal and erythema standards for a positive** response varied among the hospitals. (due to degranulation of local mast cell at site of injection)



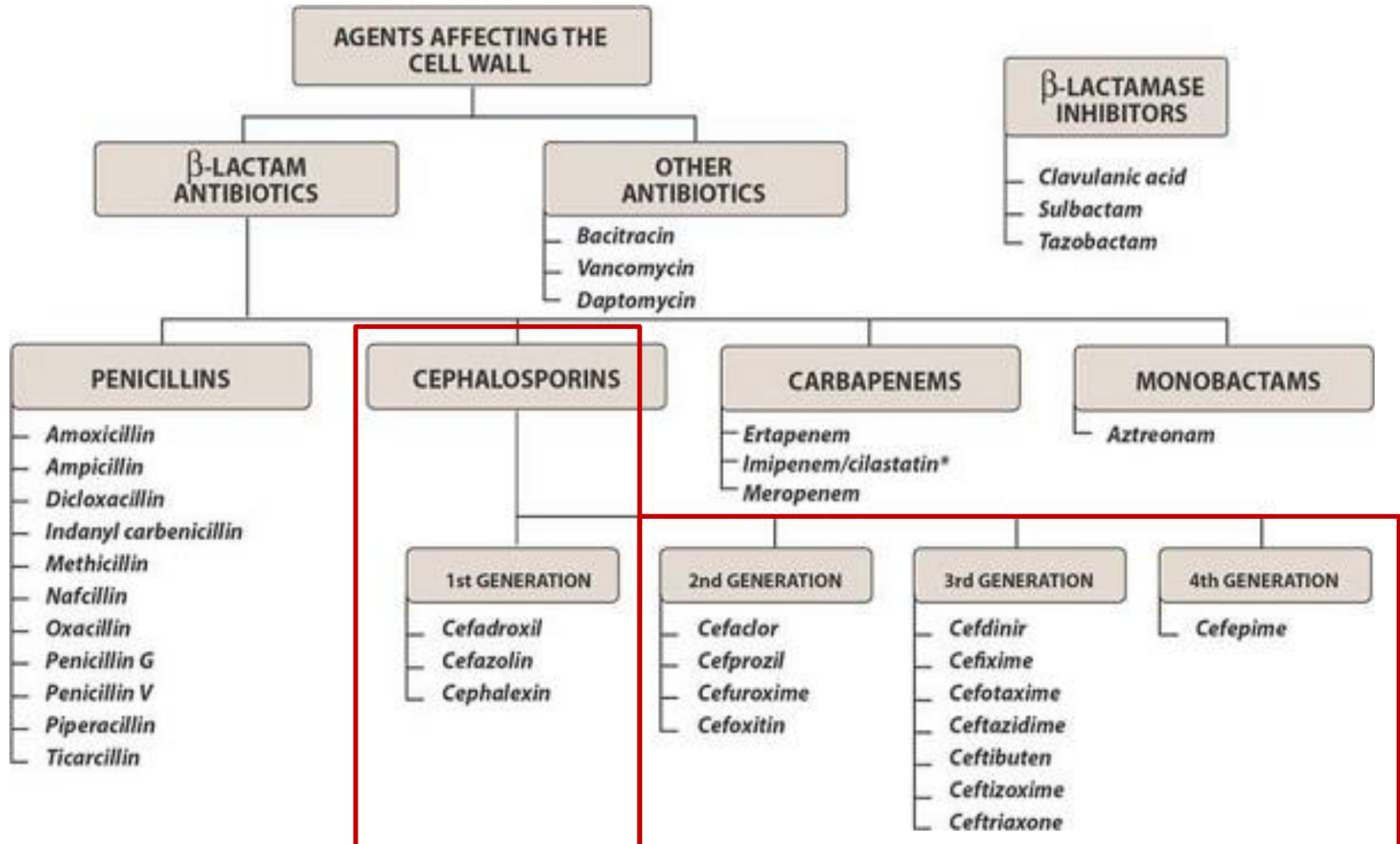
✓ **Positive skin test for specific IgE (> 3-5 mm diameter)**





Assume complete
cross-allergenicity
between individual
penicillins.

Bacterial cell wall synthesis inhibitors



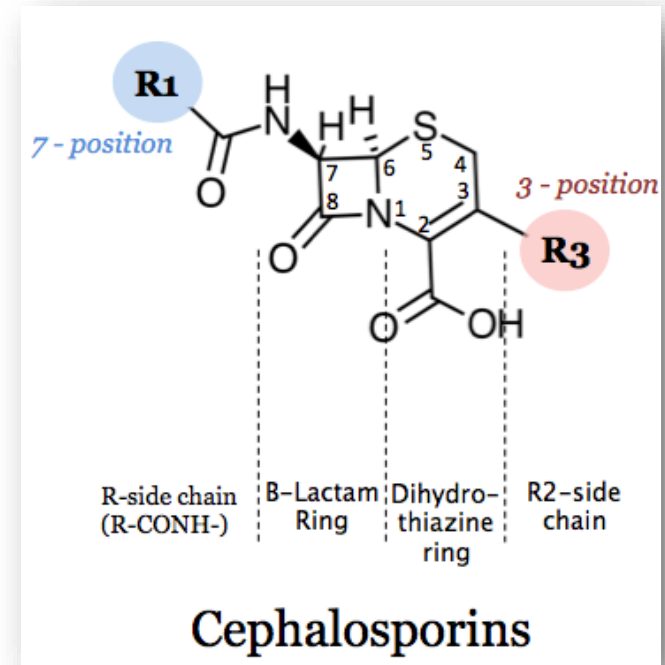
Cephalosporins (1945)

- β -lactam antibiotics (similar to penicillins)- (bind to PBP*)
- Act by bacterial cell wall synthesis inhibition

1. More resistant to β -lactamase.

2. Broad spectrum

- Bactericidal
- Inactive against: "LAME + C":
 - *Listeria monocytogenes*
 - *Legionella*
 - Atypical (*Chlamydia*, *Mycoplasma*)
 - MRSA
 - Enterococci
 - *Clostridium difficile*



- They are commonly used in surgical procedures to reduce post operative infections risk.

Cephalosporins Subgroups

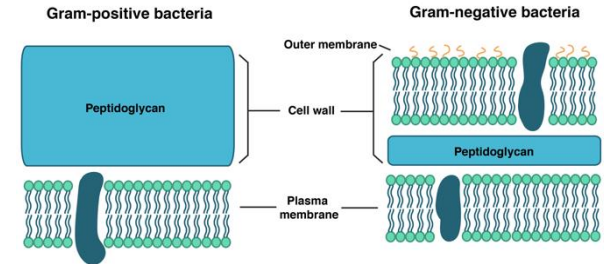
❖ FIRST-GENERATION

- Cefazolin

- Cephalexin (p.o.)

- Cephradine

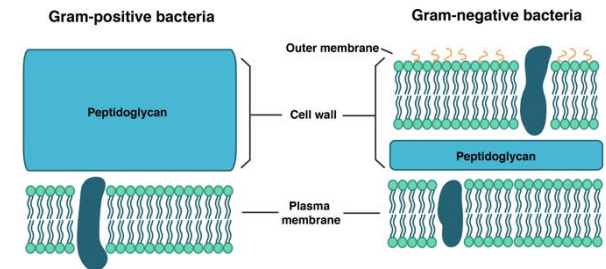
- Active against G+ organisms (S.aureus, S.pneumoniae, S.pyogenes, S.epidermidis)
- Active against streptococcal pharyngitis
- **Commonly used in surgical prophylaxis**
- None enter CNS
- Moderate activity against G- organisms (E.coli and K. pneumoniae,..)



29
Streptococcal pharyngitis

❖ SECOND-GENERATION

- Cefaclor
- Cefuroxime
- Cefoxitin
- Cefprozil



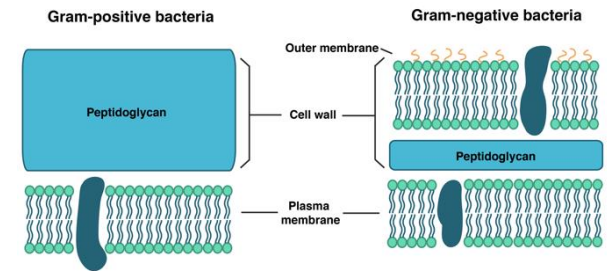
- **Mainly effective against G- organisms.**
- (H.influenza, Moraxella catarrhalis, S. pneumoniae)
- Active against intestine anaerobes (Bacteroides fragilis)
- ❌ **Not active against P.aeruginosae**
- Drugs in this group are used mainly for URTIs (acute otitis media, sinusitis, laryngitis) and LRTIs (chronic bronchitis, COPD)
- **Do not enter CNS**
- **Moderate activity against G+ organisms.**



acute otitis media

❖ THIRD-GENERATION

- Cefotaxime
- Ceftriaxone
- Ceftazidime
- Cefoperazone
- Cefixime (p.o.)



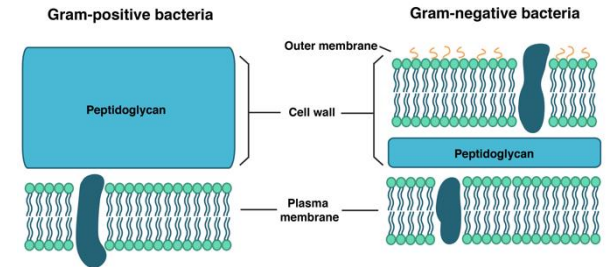
- **Most** active **against G- organisms** (H.influenza, **N. gonorrhea**, E.coli, **N.meningitidis**, M. catarrhalis, Klebsiella spp and **P.aeruginosae**).
- **They penetrate BBB** except (Cefixime and Cefoperazone).
- Less active against G+ organisms



❖ FOURTH-GENERATION

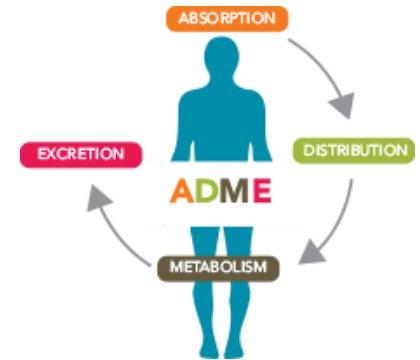
-Cefipime (IV)

- Wider spectrum that includes G+ activity and G- cocci activity, plus many G- rods.
- Is active against *P. aeruginosae*
- They penetrate BBB
- Resistant to most beta-lactamases



Cephalosporins: Pharmacokinetics

- Cephalosporins are **mostly administered parenterally** but some are available for oral use.
- **First** and **Second** generations CANNOT cross **BBB**, however, **Third** and **Fourth CAN** cross (except **cefoperazone**)
- Relatively lipid insoluble (like penicillins)
- Major elimination is **renal excretion**.
- $T_{1/2} = 30-90$ min.
- Ceftriaxone has long $T_{1/2} = 4-7$ h
- **Probenecid** inhibits their elimination and increase their blood levels (except. Ceftazidime and Cefoperazone)



Cephalosporins: clinical uses

M: Meningitis

O: Osteomyelitis

S: Skin infection

T: Typhoid Fever

U:UTI

R:RTI

G:Gonorrhea

E: ENT infection

N:Intestinal Infections

T:Topical

➤ Adverse effects

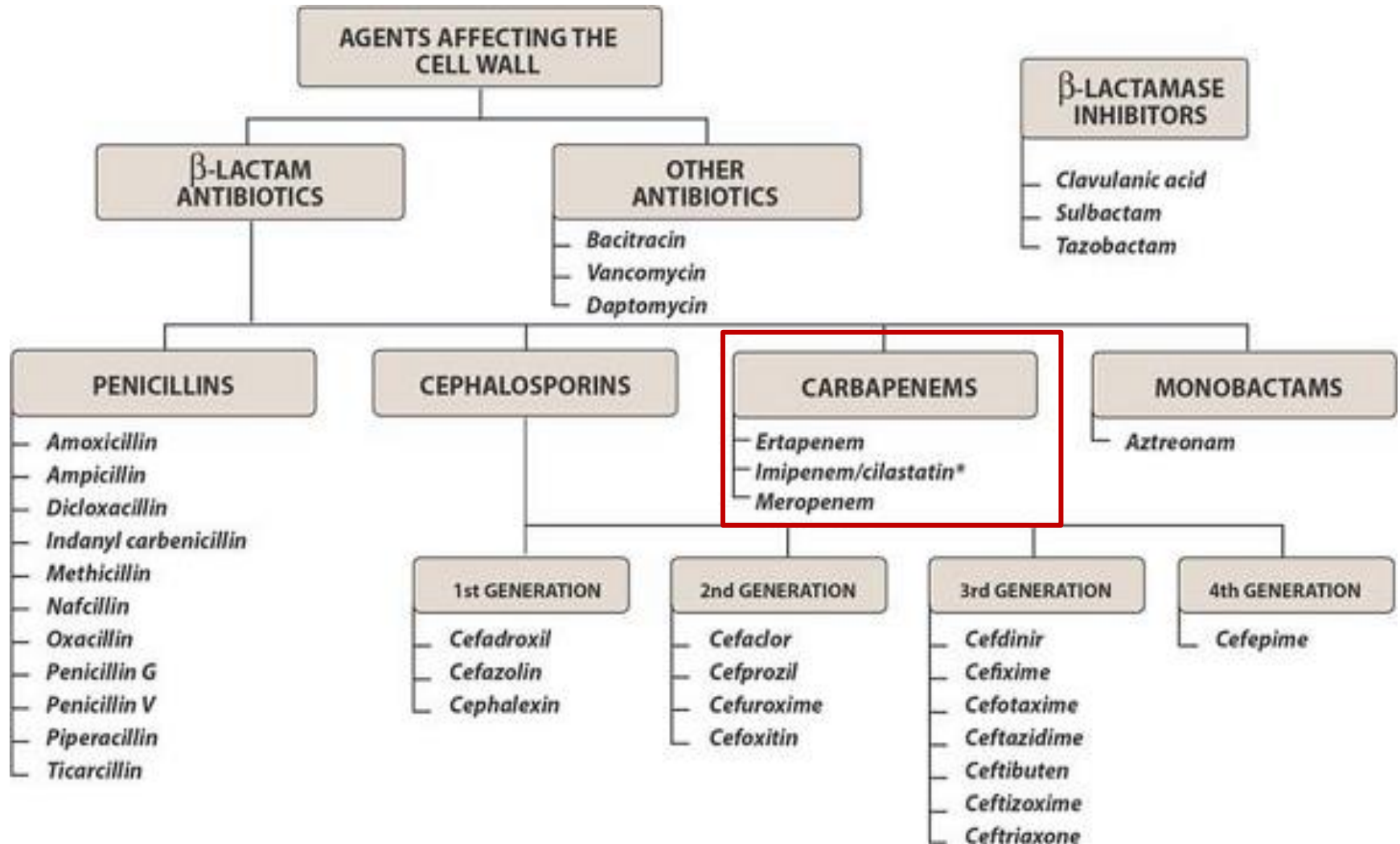
- Similar to Penicillins

Q: Patient has allergic to Penicillins can we use Cephalosporins?

- About 5-10 % show cross-sensitivity between penicillins and cephalosporins.
- Therefore, patients with history of anaphylaxis to penicillins **Will not be** treated with a cephalosporin.



Bacterial cell wall synthesis inhibitors



CARBAPENEMS

- Imipenem/ Cilastatin (I.V infusion) (GPCs)
 - Meropenem (I.V bolus or infusion) (GNRs)
 - Ertapenem (I.V)
- β -lactam antibiotics (similar to penicillins and cephalosporins)- (bind to PBP*)
 - Act by bacterial cell wall synthesis inhibition
 - They have wide activity against aerobic and anaerobic G+ and G- organisms (including Pseudomonal infections, Enterobacter, Citrobacter (*Infant meningitis*) and Serratia species). [ACTIVE AGAINST ALMOST EVERYTHINGS] Last resort
 - Pass BBB

*PBP- Penicillins binding protein



CARBAPENEMS: Pharmacokinetics

■ Imipenem / **Cilastatin**

- Is not absorbed orally (**I.V infusion**)
- Inactivated by renal dehydropeptidase-I enzyme - administered with cilastatin (inhibitor of this enzyme), this lead to **increase the plasma levels of drug**
- Excreted primarily by kidney
- $T_{1/2} = 1 \text{ h}$

➤ **Clinical uses:**

- Used in **empiric therapy** for infections that required multiple antibiotics (**Nosocomial infections**)
- **MRSA are resistant.**



➤ Adverse effects

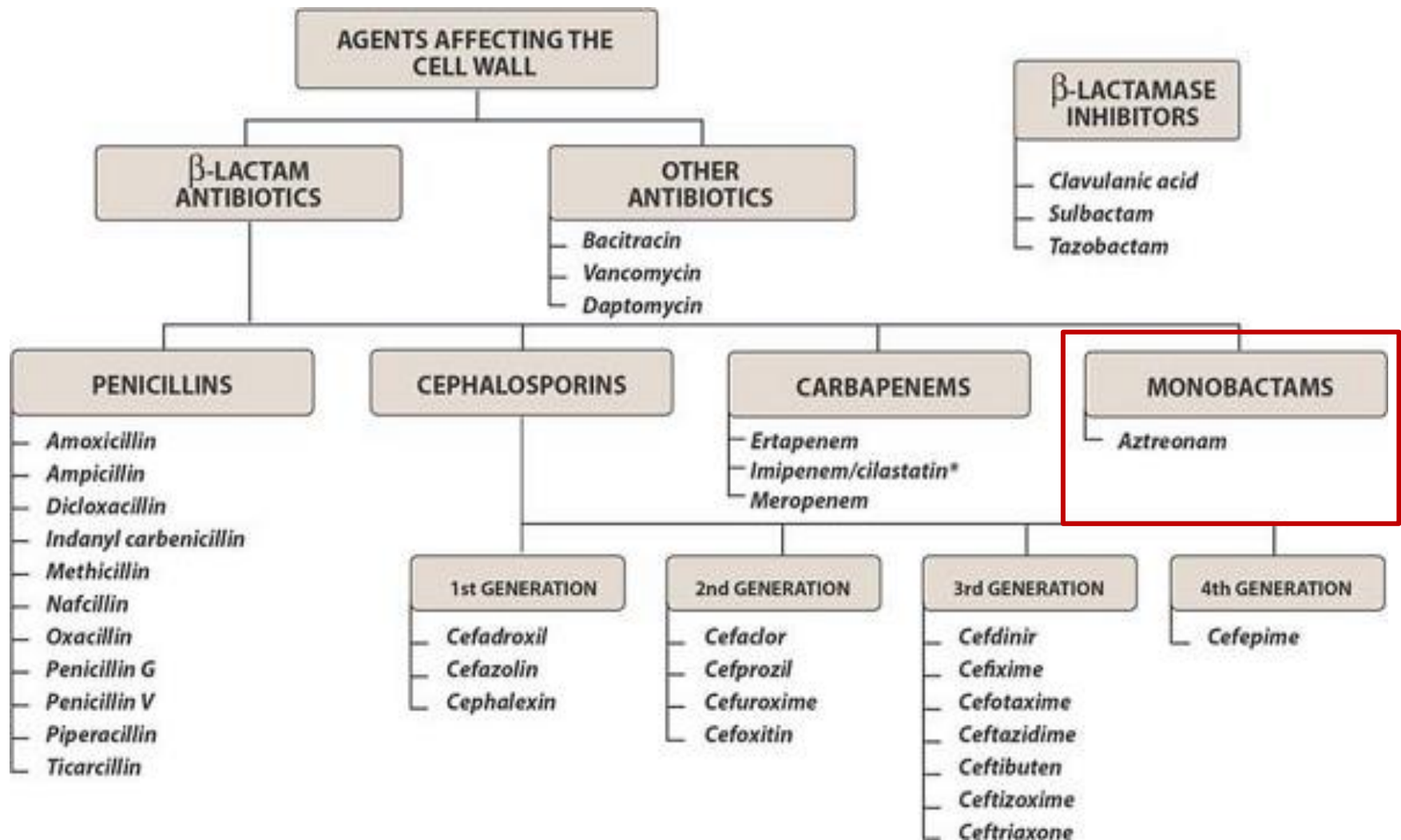
- CNS toxicity at high plasma levels (confusion, seizures)
- ✓ Meropenem :
 - Similar to Imipenem but not metabolized by renal dehydropeptidase-I and **less likely to cause seizures**.

✓ Meropenem better than Imipenem



Q: Can patient allergic to Penicillins and Cephalosporins use Carbapenems?

✗ Patients allergic to **penicillins** and **cephalosporins** may be allergic to **Carbapenems**



MONOBACTAMS

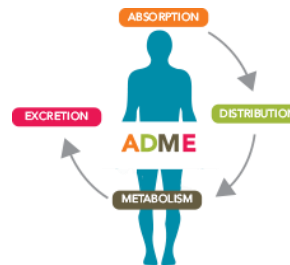
- Aztreonam (AztrEonam)

- Similar to other β -lactam drugs.
- Active against G- aerobic rods organisms (*Klebsiella*, *Pseudomonas*, *N.meningitidis*, *H.influenzae*, *enterobacteriaceae*, *N. gonorrhea* and *Serratia*)
- Not cross BBB



➤ Pharmacokinetics:

- Administered as I.V and I.M
- Limited penetration CSF
- Excreted primarily by kidney
- $T_{1/2} = 2 \text{ h}$



A E = aerobic

n = negative

➤ Adverse effects

- Similar to other β -lactams

➤ Clinical uses:

- Alternative choice for penicillins (**Piperacillin**) and cephalosporins (**ceftazidime**) **allergic patients** for G- infections.
- Alternative choice to aminoglycosides, in particular, elderly renal impairments patients



Q: Can patient allergic to Penicillins and Cephalosporins use Aztreonam?

✓ Patients allergic to **all β - lactam antibiotics** can take **Aztreonam**



What antibiotics can be used in penicillin allergy?

Most authorities recommend **avoiding cephalosporins in patients allergic to penicillins:**

G+ organisms —————> Consider **Macrolides** (e.g. [Erythromycin](#))
or **Quinolones** (e.g. [ciprofloxacin](#),
[levofloxacin](#))
or **Vancomycin** (serious infection)

G- organisms rods —————> Consider **Aztreonam**



II- GLYCOPEPTIDES

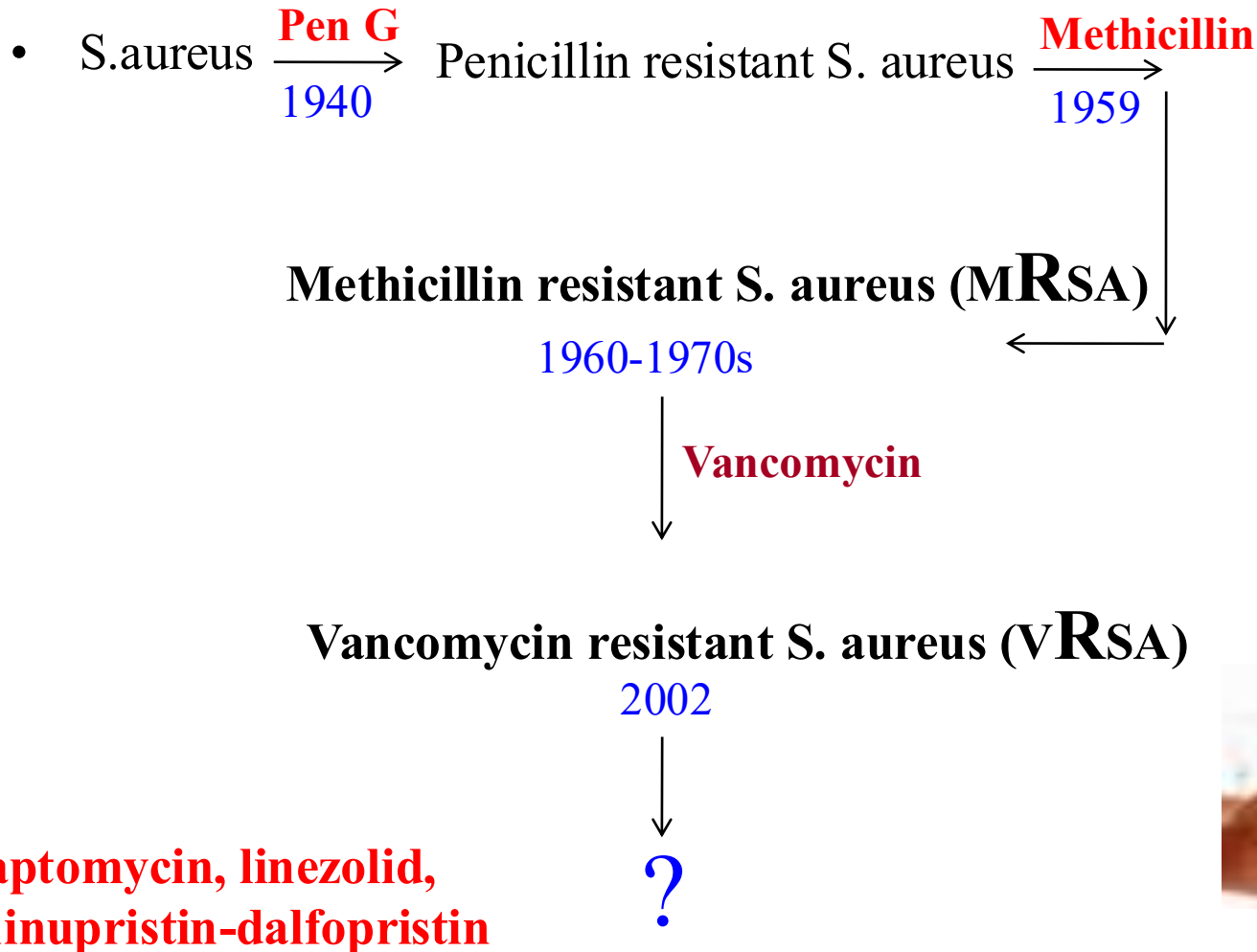
- Vancomycin ; Teicoplanin
- Bacitracin

Vancomycin:

- Inhibits cell wall synthesis by inhibits transglycosylation reaction.
- Bactericidal glycoprotein
- Active against aerobic G+ organisms, especially (Staphylococci including β -lactamase producing and **MRSA**)
- Active against anaerobic G+ organism (C.difficile)
- Not effective against G- organisms because its molecule is very large to penetrate the outer membrane of G- organisms.

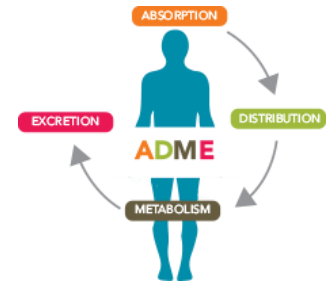


Historical perspective



Vancomycin: pharmacokinetics

- Is not absorbed from the GIT (Given orally in antibiotic-associated **colitis**, in particular by **c.difficile**)
- Its given I.V infusion
- Widely distributed (even CSF in inflammation)
- Excreted unchanged in the **urine**
- Dosage modification is necessary in patients with renal impairment.
- $T_{1/2} = 8 \text{ h}$



➤ Clinical uses:

- Staphylococcal infections caused by **MRSA** or **MRSE**.
- Prophylaxis for common surgical procedures in hospitals with high rates of infections due to MRSA or MRSE.
- Antibiotics-associated colitis (D.O.C is **Metronidazole**)
- Vancomycin-resistant staph then use **Daptomycin, linezolid**



➤ Adverse effects

- Chills, Fever
- Phlebitis at the site of injection.
- Ototoxicity , Nephrotoxicity (high conc especially with aminoglycosides)
- **“red man syndrome”** flushing (due to rapid I.V infusion- histamine release)
- Should be over at least 60 min



Bacitracin:



- Inhibits bacterial cell wall synthesis
- **Most active against G+ organisms** (Staphylococci including β -lactamase producing and MRSA)

➤ Clinical uses:

- Because of its nephrotoxicity, **Not used systemically**.
- Used **topical** for skin infections only.



	1 ST GENERATION	2 ND GENERATION	3 RD GENERATION	4 TH GENERATION
	Cephalexin Cephhradine Cefadroxil	Cefoxitin Cefuroxime Cefaclor	Cefotaxime Cefoperazone Ceftriaxone	Cefipime
Spectrum	Mainly Gram +ve spp. and few Gram -ve bacilli	Maintain Gram +ve coverage and enhanced Gram -ve coverage	Excellent Gram -ve coverage (including <i>Pseudomonas</i>)	Wide Gram +ve and Gram -ve coverage including <i>Pseudomonas</i>
β-lactamase susceptibility	Yes	Yes	No	No
Penetration to CSF	No even in meningitis	No (except <i>cefuroxime</i>)	Yes (except <i>cefoperazone</i>)	Yes
Cross allergy with penicillin	High	Moderate	Low	Low

